



The questions on the following page are an exercise to help us see applicants' approaches to quantitative analysis. Please answer the questions by creating and emailing us a **PowerPoint slide deck which summarizes your analysis and results, and any files (Excel, python, R, etc.) used for calculations**. These questions should take a maximum of 4 to 5 hours—please don't feel it necessary to spend longer than that. We hope you'll find working through these questions to be an interesting example of applying analysis to energy planning.

Action needed: Please reply to the email within the time frame requested. We will review your materials and will be in touch with you with status and next steps in the 1-2 weeks following receipt of your materials.

How we use this assessment: As we review candidates' responses, we are looking for indications of following qualities:

- + Initiative – The calculation steps required for each question are intentionally left somewhat ambiguous. This is consistent with most work at E3, where clients or project manager may have important guiding questions, but do not directly prescribe a single modeling approach. We also recognize (and take into account) that many candidates have not yet had exposure to this subject matter. Feel free to use any internet resources (including energy planning guides) to help with the assessment.
- + Reasonableness – We are primarily concerned with the reasonableness of assumptions, approaches, and results. We are not looking for a single particular answer.
- + Clarity and aesthetics – The slides and calculations should be easy to follow and should show professional polish. Communication is an important aspect of our consulting work. For the slide deck, please prepare materials which are appropriate as a standalone communications document.
- + Simplicity – While precision is good, we are also looking to see that the time you have spent on each component of the calculation is informed by its importance to the overall result. The reason is twofold: first, it is not our intention that this assignment should take more than a few hours; second, prioritizing where to spend time in a calculation is critical to doing effective consulting work at E3.

Please don't hesitate to contact us with any needed questions or clarifications.

Kind Regards,

E3 Recruiting Team

Applicant Exercise:

Green Charge Ventures (GCV), a renewable energy developer, has recently approached E3 about performing a valuation on a potential solar + storage installation project. GCV has acquired a parcel of land in New York City and would like to investigate the cost-effectiveness of different configurations of solar + storage, including standalone solar, standalone storage, or a system that combines solar and storage. To help you answer their questions, they have provided data on solar + storage systems they are considering in a separate Excel attachment. This attachment includes information on the size and costs of the solar + storage configurations, as well as a normalized solar production shape.

Questions for Applicants:

1. To start, GCV would like to investigate the historical energy prices in New York City. The project is slated to be in the “N.Y.C.” zone of the New York ISO (NYISO) service territory. GCV has provided a set of historical prices and would like to understand the value streams associated with them
 - a. Please provide the monthly averaged historical day-ahead energy prices for 2023.
 - b. Discuss and explain which periods of the year will be most valuable for the proposed solar installation.
 - c. Discuss and explain which periods of the year will be most valuable for the proposed storage installation.
2. After determining the historical prices, GCV would like to analyze potential revenue streams from their suite of proposed projects. (Note: rigorous revenue analysis can be accomplished using both price streams or one of your choosing. Rationale for selection can optionally be included in accompanying slides)
 - a. Using the 2023 historical energy prices, please calculate a projected backcast 2023 revenue for the GCV solar system.
 - b. Using the 2023 historical energy prices, please calculate a best estimate for the projected backcast 2023 revenue for the GCV standalone storage system. (Note: This section (part b) is not intended to take more than one hour. Please make any simplifications necessary and state assumptions)
 - c. Using the 2023 historical energy prices, please calculate a best estimate for the projected backcast 2023 revenue for the GCV combined solar + storage system. GCV would particularly like to understand how solar and storage would interact if paired together and how revenues might increase or decrease compared to standalone projects. (Note: This section (part c) is not intended to take more than one hour. Please make any simplifications necessary and state assumptions)
3. After seeing the set of historical revenues provided, GCV is cautiously optimistic, but would like to understand how the revenue streams will evolve throughout the upcoming decade
 - a. Please provide an explanation on how you believe the energy price revenue stream for each project will change over time, and reasons why.
 - b. GCV has proposed a potential follow-up project with E3 for a more detailed investigation of the revenue analysis. Please prepare a short summary of the approach you would take to prepare a more complete revenue forecast if given an extra 10 hours of work.
 - c. Which, if any, of the three proposed installations would you recommend for GCV installation?